

# CPU-GPU Collaborative Compact Incremental Computation

## Insert repair stage

Affected-Vertex  
Repair

Construct  
Compact Worklist

Insertion  
Relaxation

## Adaptive Workload mode

Ordered  
Propagation  
(large-diameter graph)

Shared batch  
preparation  
(large batch update)

## Deletion Repair

Dependency  
Invalidation

Construct  
Compact Worklist

Affected-Vertex  
Repair

Phase-wise  
Convergence

Previous Result:  $R_t$

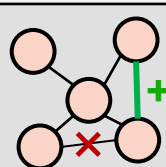
Result:  
 $R_{t+1}$

Delete Edges

Request Precursor

Compact Precursor

Graph Update  
 $G_t + \Delta G_t \rightarrow G_{t+1}$



Source-local Updates

Unified Change



Delta List

Asynchronous  
Flash

Base Reverse Index

GPU Descriptor View

## Heterogeneous Source-Isolated Graph Organization

CPU-Initiated  
Transfer

GPU-Initiated  
Transfer